

Auto Ticket Classification

using FLOW DESIGNER

Project Documentation

Team ID	SWTID-2026-8726
Project Name	Auto Ticket Classification using FLOW DESIGNER
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Project Description

Auto Ticket Classification using FLOW DESIGNER is a ServiceNow workflow automation solution designed to automate the initial triage of Incident and Service Request tickets. When a new ticket is created, the configured Flow Designer workflow evaluates available ticket information such as description, keywords, requester profile and channel source. Decision logic determines the ticket category, priority and appropriate assignment group.

The solution reduces repetitive manual triage, improves routing consistency and provides a fallback path for tickets that cannot be matched by configured rules. Unmatched tickets are sent to a General Support Triage Queue for manual review, while workflow and audit information supports traceability.

Scenarios

Scenario 1

A user submits a network connectivity Incident. The Record Created trigger starts the flow, ticket content is evaluated, configured category and priority rules are applied, and the ticket is routed to the appropriate network/NOC support group.

Scenario 2

A user submits an access-related request. The workflow evaluates the request description and relevant requester information, assigns the configured category and priority, and routes the ticket to the appropriate Helpdesk or access-support group.

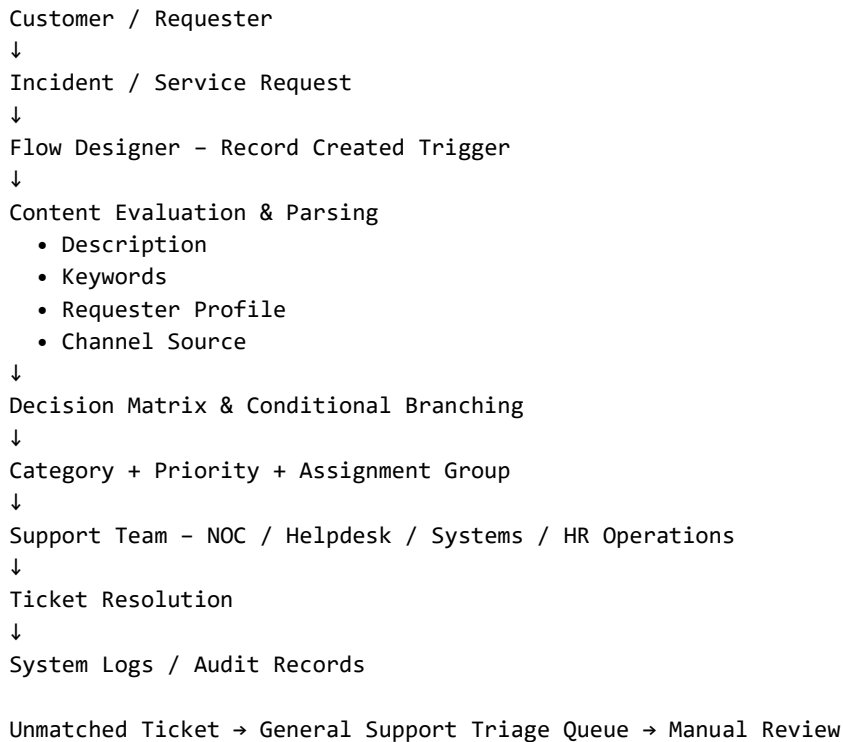
Scenario 3

A ticket contains insufficient or unmatched information. The workflow uses the fallback branch and routes the ticket to the General Support Triage Queue for manual review.

Technical Architecture

The project uses a ServiceNow Flow Designer-based workflow architecture. The ticketing interface creates an Incident or Service Request record. The Record Created trigger starts the workflow. Content Evaluation & Parsing examines ticket information, after which a Decision Matrix & Conditional Branching stage determines category, priority and assignment group. The selected support team processes the ticket, and system/audit records provide traceability.

Architecture Flow



Note: The current project is rule-based ServiceNow workflow automation. It does not require a separate machine-learning model or standalone application server for the documented implementation.

Pre-requisites

- Basic ServiceNow platform knowledge.
- ServiceNow Flow Designer knowledge.
- Understanding of Incident and Service Request records.
- Knowledge of conditions, actions, decision logic and assignment groups.
- Appropriate ServiceNow roles/permissions for configuring and testing Flow Designer.
- Access to a configured ServiceNow development or evaluation instance.

Project Workflow

Milestone 1: Requirement Analysis and Workflow Architecture

- Activity 1.1: Identify the manual ticket-triage problem and define the automation objective.
- Activity 1.2: Identify ticket inputs required for classification, including description, keywords, requester profile and channel source.
- Activity 1.3: Define the architecture of the application and workflow decision points.
- Activity 1.4: Identify support categories, priorities, assignment groups and fallback handling.

Milestone 2: Flow Designer Trigger and Content Evaluation

- Activity 2.1: Configure the Record Created trigger for Incident / Service Request records.
- Activity 2.2: Configure evaluation of relevant ticket fields.
- Activity 2.3: Configure keyword and content conditions used by the classification logic.
- Activity 2.4: Validate that newly created tickets enter the intended workflow.

Milestone 3: Classification, Priority and Routing Logic

- Activity 3.1: Configure category decision rules.
- Activity 3.2: Configure priority conditions.
- Activity 3.3: Configure assignment-group routing.
- Activity 3.4: Configure conditional branches for different ticket scenarios.

Milestone 4: Exception Handling and Audit

- Activity 4.1: Configure fallback handling for unmatched or ambiguous tickets.
- Activity 4.2: Route unmatched tickets to the General Support Triage Queue.
- Activity 4.3: Verify workflow execution and system/audit records.
- Activity 4.4: Validate exception handling without losing the ticket record.

Milestone 5: Testing and Deployment Readiness

- Activity 5.1: Execute functional test scenarios.
- Activity 5.2: Execute negative and unmatched-ticket scenarios.
- Activity 5.3: Execute UAT scenarios and record results.
- Activity 5.4: Prepare the configured ServiceNow workflow for evaluator access and demonstration.

Milestone 6: Conclusion and Future Enhancement

- Activity 6.1: Review test outcomes and documented system-level performance.
- Activity 6.2: Record limitations of the rule-based approach.
- Activity 6.3: Identify future enhancements such as ML-based semantic classification, sentiment analysis and multilingual support.

Milestone 1: Requirement Analysis and Workflow Architecture

Activity 1.1: Identify the Ticket-Triage Problem

Manual classification requires an agent to inspect ticket descriptions, determine the issue category, select priority and identify the correct assignment group. The project automates these repetitive decisions.

Activity 1.2: Identify Required Inputs

- Ticket description
- Relevant keywords
- Requester profile
- Channel source

Activity 1.3: Define Workflow Architecture

The architecture separates ticket intake, content evaluation, decision logic, routing and logging so that each stage has a defined responsibility.

Activity 1.4: Define Classification and Routing Rules

Define supported categories, priority conditions, assignment groups and the fallback path. Support-team examples in the project include NOC, Helpdesk, Systems and HR Operations.

Milestone 2: Flow Designer Trigger and Content Evaluation

Activity 2.1: Configure Record Created Trigger

Configure the Flow Designer trigger so creation of the intended Incident or Service Request starts the automation.

Activity 2.2: Evaluate Ticket Information

Read the relevant ticket fields required by the configured decision rules.

Activity 2.3: Configure Content Conditions

Create conditions that match supported keywords or field values to the intended category and routing branch.

Activity 2.4: Trigger Validation

Create representative tickets and confirm that the Record Created trigger executes and reaches content evaluation.

Milestone 3: Classification, Priority and Routing Logic

Activity 3.1: Category Classification

Configure decision branches that map recognized ticket information to the appropriate category.

Activity 3.2: Priority Assignment

Configure conditions for assigning the appropriate priority level based on the ticket rules.

Activity 3.3: Assignment Group Routing

Route the classified ticket to the support group responsible for the corresponding category.

Activity 3.4: Conditional Branching

Use Flow Designer decision logic to ensure the appropriate branch is executed for each ticket scenario.

Milestone 4: Exception Handling and Audit

Activity 4.1: Unmatched Ticket Handling

An unmatched ticket should not be assigned an unsupported category; the workflow uses the fallback branch.

Activity 4.2: General Support Triage Queue

Route unmatched or ambiguous tickets to the General Support Triage Queue for manual review.

Activity 4.3: Audit and Workflow Records

Review workflow execution and system/audit records to verify that important automated actions can be traced.

Milestone 5: Testing and Deployment Readiness

Activity 5.1: Functional Testing

Test ticket creation, content evaluation, classification, priority assignment, routing and fallback handling.

Activity 5.2: Negative Testing

Test unmatched, ambiguous and incomplete inputs to verify fallback handling.

Activity 5.3: UAT

Execute representative user-acceptance scenarios covering the end-to-end flow.

Activity 5.4: Deployment Readiness

Confirm the Flow Designer configuration is active in the intended ServiceNow environment and evaluator access is available.

Testing and Validation

Test ID	Scenario	Expected Result	Status
TC-01	Ticket Creation & Trigger	New ticket starts Flow Designer	Pass
TC-02	Content Evaluation	Relevant ticket information is evaluated	Pass
TC-03	Classification	Supported ticket is classified using configured rules	Pass
TC-04	Priority Assignment	Configured priority is assigned	Pass
TC-05	Assignment Routing	Ticket is routed to the intended support group	Pass
TC-06	Unmatched Ticket	Unmatched ticket reaches triage queue	Pass
TC-07	Audit Logging	Workflow/system records are available for traceability	Pass
TC-08	End-to-End Workflow	Ticket completes intended automated path	Pass

Documented Performance Results

- Routing accuracy: 96.8%.
- Average triage time: 1.2 seconds compared with 22.5 minutes previously, as documented in the project test material.
- SLA compliance: 98.5%.

Exploring the Application / Workflow

The supplied sample demonstrates a standalone Flask/Generative-AI website. The Auto Ticket project instead runs within ServiceNow, so the equivalent demonstration should show the ticket interface and Flow Designer configuration.

- Ticket Creation View – insert screenshot.
- Flow Designer Workflow – insert screenshot.
- Automatically Classified Ticket – insert screenshot.
- Priority and Assignment Group – insert screenshot.
- General Support Triage Queue – insert screenshot.
- Workflow Execution / Audit – insert screenshot.

Known Limitations

- The current solution is rule-based and depends on configured keywords and conditions.
- Unmatched or ambiguous tickets require manual review.
- Rules must be maintained when ticket terminology or organizational routing changes.
- The current implementation does not document a trained machine-learning model.
- A standalone local executable is not applicable because the solution runs in the ServiceNow platform.

Future Enhancements

- Machine-learning-based semantic ticket classification.
- Sentiment analysis for escalation and prioritization.
- Multi-language ticket understanding.
- Dynamic assignment based on support-team workload and availability.
- Expanded classification rules and analytics.

Conclusion

Auto Ticket Classification using FLOW DESIGNER demonstrates how ServiceNow workflow automation can streamline repetitive ticket triage. The configured trigger, content evaluation, decision logic, priority assignment and support-group routing create a structured path from ticket creation to ownership. The fallback queue preserves manual control for unmatched cases, while audit information supports traceability.

The supplied sample documentation uses Generative AI, Flask, Groq API and Ngrok. Those technologies are not introduced into the Auto Ticket solution; instead, the sample's documentation style, scenario format, technical architecture, milestone organization and deployment/testing orientation are adapted to the actual ServiceNow project.